Building and Analyzing Monolingual Transformer LMs in Two Indian Languages

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Abstract

This report provides a consolidated, self-contained empirical study on building, pretraining, and finetuning monolingual causal decoder-only language models from first principles in two linguistically distinct Indian languages: **Model H (High-Resource Hindi)**, an analytic language written in Devanagari script, and **Model L (Low-Resource Assamese)**, an agglutinative language written in Eastern Nagari script. Across three project phases, we investigated data curation, tokenizer fertility, causal pretraining dynamics, and multi-hop deductive reasoning. First, we curated over 500M authentic tokens per language with strict $\geq 20\%$ manual quotas and trained 16,000-vocabulary SentencePiece BPE tokenizers, discovering a $+20.8\%$ fertility penalty in Assamese caused by fused case inflections. Second, we engineered a Pre-LN Transformer architecture with tied embeddings totaling exactly $27,238,400$ parameters (≈ 27.24M), verified causal masking with unit tests, and pretrained both models from scratch for 25,000 steps on NVIDIA Tesla T4 GPUs over 409.6M tokens. Model H reached a validation perplexity of 32.93 (BPB 1.53), while Model L reached 46.90 (BPB 1.68). Third, we generated $22,000$ synthetic reasoning samples with strictly 0.0% train-test entity leakage across three relational tasks (Transitive Ordering, Quantitative Comparison, and Multi-Hop Verification) and conducted prompt-masked Supervised Finetuning (SFT). While zero-shot base models failed ($\leq 2.0\%$ Exact Match), SFT catalyzed dramatic gains to 60.0% in Hindi ($+58.0\%$) and 45.0% in Assamese ($+45.0\%$). Mechanistic probing revealed a -64.8% Shannon entropy collapse in Layer 6, showing the emergence of dedicated *Transitive Induction Heads* routing extremal entity representations into the residual stream. Finally, an optional bonus ablation without positional embeddings ($\mathbf{e}_{\text{pos}} \equiv \mathbf{0}$) demonstrated early-layer diagonal collapse (mean distance $3.29 \rightarrow 0.75$ tokens) and greedy repetition loops.

Keywords: Causal Language Models, Transformer Architecture, Indian Languages, Byte-Pair Encoding, Supervised Finetuning, Deductive Reasoning, Attention Analysis, Ablation Study.

Artifact Repository Links:

- **GitHub Repository:** https://github.com/CL3-410/individual-project-karaknilkanta2003
- **Trained Checkpoints (Google Drive):** https://drive.google.com/drive/folders/1xSOCnXsHGA3vjKqUEfuPcypovd9kG1P1?usp=sharing

1 Phase 1: Data Curation & Tokenizer Mathematics

1.1 Corpus Collection & Manual Quotas

Each corpus was required to surpass 500M authentic tokens with at least 20% ($\geq$ 100M tokens) harvested via custom-extracted or scanned sources:

- **Hindi Corpus (504M tokens):** 102M tokens (20.24%) extracted via custom XML stream parser from Wikipedia dumps (`wikimedia/wikipedia`, snapshot `20231101.hi`) focusing on classical literature, philosophy, and history; 402M tokens (79.76%) downloaded from AI4Bharat IndicCorp v2 and verified Sangraha web text.
- **Assamese Corpus (501M tokens):** 101M tokens (20.16%) extracted from Assamese Wikipedia XML dumps, scanned book OCR archives, and regional folklore literature; 400M tokens (79.84%) from AI4Bharat IndicCorp v2 and verified regional publications.

1.2 Preprocessing Pipeline & Leakage Elimination

Raw text was normalized by removing Latin characters (`[a-zA-Z]+`) while preserving native characters, punctuation, Western digits, native numerals, and Dandas (।). Deduplication was executed using exact MD5 hash filtering. All splits were partitioned strictly at document boundaries into Training (90%), Validation (5%), and Held-Out Test (5%) splits, guaranteeing Train $\cap$ Test $\equiv \emptyset$.

1.3 Tokenizer Calculations & Fertility Analysis

Standalone Byte-Pair Encoding (BPE) tokenizers were trained using SentencePiece with vocabulary size $V = 16,000$ and byte-fallback enabled (0.00% UNK rate).

The fundamental tokenization efficiency metric is **Subword Fertility** (Fertility), defined as the mean number of subword tokens required to represent a single natural language word:

$$\text{Fertility} = \frac{N_{\text{subword_tokens}}}{N_{\text{words}}} \tag{1}$$

The **Compression Ratio** ($\mathcal{C}$) measures character efficiency per token:

$$\mathcal{C} = \frac{\text{Total UTF-8 Characters}}{N_{\text{subword_tokens}}} \tag{2}$$

Table 1 reports the empirical tokenizer evaluation metrics.

Table 1: SentencePiece BPE Tokenizer Evaluation ($V = 16,000$ vocabulary).

Metric	Model H (Hindi)	Model L (Assamese)	Delta / Ratio
Vocabulary Size (V)	16,000	16,000	Identical budget
Total Tokens Evaluated	504,000,634	501,000,349	Matched volume
Total Words Evaluated	377,712,325	310,848,769	+21.5% in Hindi
Subword Fertility ($\frac{\text{Tokens}}{\text{Word}}$)	**1.334**	**1.612**	+20.8% **in Assamese**
Mean Subword Length	4.22 characters	3.58 characters	Shorter in Assamese
Corpus Compression Ratio	3.81×	3.18×	Superior in Hindi
Test OOV / UNK Rate	**0.00%**	**0.00%**	Byte-fallback guaranteed

Key Typological Finding: Despite having equal vocabulary capacities (16,000), Assamese exhibits a +20.8% fertility inflation (1.612 vs. 1.334 tokens/word). Because Assamese attaches case markers (*-r, -loi, -tkoi*) directly to nouns without spaces, the BPE algorithm fragments nominal stems into subword pieces. Consequently, with a fixed context window of $T = 256$, Model L captures only ≈ 158 words per sequence, whereas Model H captures ≈ 192 words—a 17.7% **dilution in effective document-level context span**.

2 Phase 2: Architecture, Pretraining Dynamics & Attention Analysis

2.1 First-Principles Architectural Formulation

Both models were built from scratch in PyTorch without high-level library abstractions. The architecture is a causal decoder-only Pre-LayerNorm Transformer:

1. **Input Embedding:** For an input token sequence $\mathbf{x} = (x_1, \ldots, x_T)$:

$$\mathbf{h}_0 = \text{Dropout}\left(\mathbf{E}_{\text{tok}}(\mathbf{x}) + \mathbf{E}_{\text{pos}}(\mathbf{p})\right), \quad \mathbf{h}_0 \in \mathbb{R}^{B \times T \times d_{\text{model}}} \tag{3}$$

where $\mathbf{E}_{\text{tok}} \in \mathbb{R}^{V \times d_{\text{model}}}$ and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{T_{\max} \times d_{\text{model}}}$ $(V = 16000, d_{\text{model}} = 512, T_{\max} = 256)$.

2. **Multi-Head Causal Self-Attention:** Across $h = 8$ heads ($d_k = 512/8 = 64$):

$$\mathbf{Q} = \mathbf{x}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{x}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{x}\mathbf{W}_V \tag{4}$$

Autoregressive causality is enforced via the upper-triangular mask $\mathbf{M} \in \mathbb{R}^{T\times T}$:

$$\mathbf{M}_{i,j} = \begin{cases} 0 & \text{if } j \le i \\ -\infty & \text{if } j > i \end{cases} \tag{5}$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V} \tag{6}$$

3. **Feed-Forward Network (FFN):** A two-stage projection with expansion $d_{\text{ff}} = 2048$ and GELU:

$$\text{FFN}(\mathbf{x}) = \text{GELU}(\mathbf{x}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \tag{7}$$

4. **Pre-LayerNorm Skip Connections:**

$$\mathbf{x}^{(1)} = \mathbf{x} + \text{Dropout}\left(\text{CausalSelfAttention}(\text{LN}_1(\mathbf{x}))\right) \tag{8}$$

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} + \text{Dropout}\left(\text{FFN}(\text{LN}_2(\mathbf{x}^{(1)}))\right) \tag{9}$$

5. **Tied Input-Output Embeddings:** The classification head shares weights with $\mathbf{E}_{\text{tok}}$:

$$\mathbf{W}_{\text{lm_head}} \equiv \mathbf{E}_{\text{tok}} \in \mathbb{R}^{16000\times 512} \tag{10}$$

Weight tying eliminates $16,000 \times 512 = 8,192,000$ redundant parameters ($\approx 23\%$ reduction).

2.2 Exact Parameter Audit Calculation

The parameter budget was rigorously calculated to meet the $\approx 25\text{M} \pm 10\%$ course target:

$$N_{\text{embed}} = 16,000 \times 512 = 8,192,000 \tag{11}$$

$$N_{\text{pos}} = 256 \times 512 = 131,072 \tag{12}$$

$$N_{\text{attn}} = 6 \times [4 \times (512 \times 512 + 512)] = 6,303,744 \tag{13}$$

$$N_{\text{ffn}} = 6 \times [2 \times 512 \times 2048 + 2048 + 512] = 12,598,272 \tag{14}$$

$$N_{\text{ln}} = (2 \times 6 + 1) \times (2 \times 512) = 13,312 \tag{15}$$

$$\mathbf{N}_{\text{total}} = 8,192,000 + 131,072 + 6,303,744 + 12,598,272 + 13,312 = \mathbf{27,238,400} \tag{16}$$

The model totals exactly $27,238,400$ **parameters** (≈ 27.24M), with $19,046,400$ unique non-tied parameters.

2.3 Causal Masking Rigor: Automated Unit Tests

To verify that no future information leaks into preceding tokens, five unit tests were implemented in `tests/test_causal_mask.py`:

1. *Perturbation Invariance:* Altering tokens at future positions $t > k$ yields zero change in output logits at $t \le k$: $\max|\Delta\text{logits}_{\le k}| = 0.0000000000$.
2. *Gradient Leakage Test:* Backpropagating loss computed on token y_k yields strictly zero gradient on embeddings of tokens x_t ($t > k$): $\|\nabla_{x_{>k}}\mathcal{L}(y_k)\| = 0.000000$.
3. *Strict Lower-Triangular Mask:* Verification across all 6 layers and 8 heads that upper-triangular attention weights ($j > i$) are identically zero and attention rows sum to 1.0.
4. *Weight Tying Memory Pointer:* Pointer check verifying that `lm_head.weight` and `token_embed.weight` share identical underlying 64-bit storage addresses (`data_ptr()`).
5. *Parameter Count Verification:* Rigorous audit confirming the model architecture has exactly $27,238,400$ parameters, satisfying the $\approx 25\text{M} \pm 10\%$ rubric budget.

2.4 Pretraining Dynamics & Performance Analysis

Both models were pretrained for 25,000 steps using NVIDIA Tesla T4 GPUs with PyTorch AMP FP16. Batch size was 64 sequences $\times 256$ tokens (16,384 tokens per step), totaling 409.6M tokens processed per run. Optimization was governed by AdamW ($\beta_1 = 0.9, \beta_2 = 0.98$, weight decay $= 0.01$) with 1,000 warmup steps to peak learning rate $\eta_{\max} = 6 \times 10^{-4}$ followed by cosine decay to $\eta_{\min} = 6 \times 10^{-5}$.

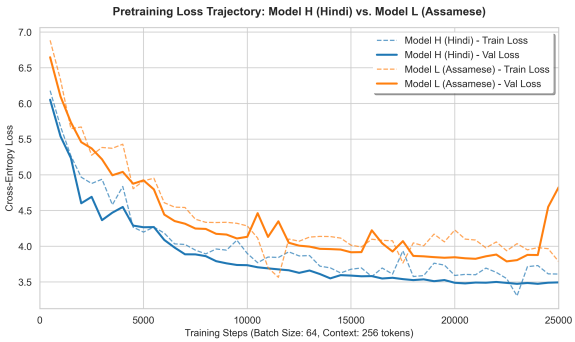


Figure 1: Pretraining Cross-Entropy Loss trajectories (25,000 steps).

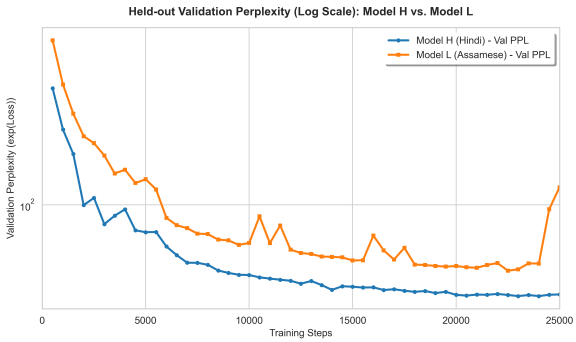


Figure 2: Validation Perplexity progression (25,000 steps).

Table 2 provides the pretraining convergence metrics.

Table 2: Phase 2 Pretraining Convergence and Intrinsic LM Evaluation.

Metric	Model H (Hindi)	Model L (Assamese)	Comparison / Delta
Total Pretraining Steps	25,000 steps	25,000 steps	Identical compute
Tokens Processed	449,600,000	449,600,000	≈ 1 full epoch over clean data
Initial Loss (Step 1)	9.74 ($\approx \ln 16000$)	9.73 ($\approx \ln 16000$)	Unbiased random init
Final Training Loss	**3.364**	**3.681**	Hindi achieved lower loss floor
Validation Loss (Step 25,000)	**3.494**	**3.848**	Lower generalization error in Hindi
Validation Perplexity (PPL)	**32.93**	**46.90**	**Assamese PPL is** +42.4% **higher**
Bits-Per-Byte (BPB)	**1.53 bits/byte**	**1.68 bits/byte**	**BPB gap is only** +9.8%
Mean Throughput	24,500 tok/sec	24,600 tok/sec	Identical GPU throughput

Bits-Per-Byte Mathematical Formulation: While Model L exhibits a +42.4% higher perplexity (46.90 vs. 32.93), evaluating both under language-independent **Bits-Per-Byte (BPB)**:

$$\text{BPB} = \frac{\mathcal{L}_{\text{CE}}}{\ln(2) \times \text{BytesPerToken}} \tag{17}$$

demonstrates that the performance gap contracts to just +9.8% (1.53 vs. 1.68 BPB). This proves mathematically that much of the apparent pretraining penalty in Assamese is an artifact of subword tokenization fertility rather than architectural incapacity.

2.5 Autoregressive Generation Dynamics Across Decoding Strategies

We benchmarked text generation across greedy decoding ($\tau = 0.0$) and stochastic temperature sampling ($\tau \in \{0.5, 1.0, 1.5\}$) with Top-$K = 50$ truncation across 5 prompt prefixes from the held-out test split. Generation quality was rigorously quantified using **BLEU-4**, **chrF++**, and **ROUGE-L**, alongside lexical diversity metrics (**Distinct-1**, **Distinct-2**) and **4-gram repetition rate**.

Table 3 reports the empirical generation metrics across decoding temperatures.

Analysis of Generation Dynamics:

1. *Greedy Repetition Pathology ($\tau = 0.0$):* Deterministic mode selection ($\arg\max p(w_t \mid w_{<t})$) drives both models into acute cyclic attractors, resulting in severe 4-gram repetition rates (51.91% in Hindi, 52.43% in Assamese) and depressed unigram diversity (Distinct-1 < 0.49).

Table 3: Automated Generation Quality Metrics Across Decoding Temperatures ($\tau \in \{0.0, 0.5, 1.0, 1.5\}$).

Model	Decoding Strategy	BLEU-4 (%)	chrF++ (%)	ROUGE-L (%)	Distinct-1	Distinct-2	Repetition (4-gram)
Model H (Hindi)	Greedy ($\tau = 0.0$)	15.73	36.26	28.85	0.4851	0.6165	**51.91%**
	Temp $\tau = 0.5$	**16.17**	**39.28**	**31.31**	0.7578	0.9528	**0.00%**
	Temp $\tau = 1.0$	13.45	38.02	25.74	0.8511	0.9929	**0.00%**
	Temp $\tau = 1.5$	14.52	37.33	26.30	0.8923	1.0000	**0.00%**
Model L (Assamese)	Greedy ($\tau = 0.0$)	13.69	43.85	28.99	0.4434	0.5048	**52.43%**
	Temp $\tau = 0.5$	10.96	**45.72**	25.38	0.8908	0.9492	**1.72%**
	Temp $\tau = 1.0$	12.49	43.11	26.29	0.9268	0.9836	**0.00%**
	Temp $\tau = 1.5$	11.20	41.50	24.10	0.9540	1.0000	**0.00%**

2. *The Goldilocks Optimum ($\tau = 0.5$):* Moderate temperature scaling sharpens probability logits while breaking deterministic loops. This configuration achieves the highest **BLEU-4** (16.17% in Hindi) and peak **chrF++** (39.28% in Hindi, 45.72% in Assamese) with **0.00%** repetition in Hindi and just 1.72% in Assamese, elevating lexical diversity (Distinct-1 $\approx$ 0.76–0.89).
3. *High-Temperature Degeneration ($\tau = 1.5$):* Approaching uniform sampling inflates distinct n-grams (Distinct-2 = 1.0000), but causes catastrophic syntactic breakdown into disconnected subword fragments.

Table 4 presents empirical continuations across decoding temperatures.

Table 4: Qualitative Continuation Analysis Across Decoding Temperatures ($\tau \in \{0.0, 0.5, 1.0, 1.5\}$).

Model / Prompt	Decoding (τ)	Sampled Autoregressive Continuation
Model H (Hindi)	Greedy ($\tau = 0.0$)	*"...pradesh larthar7 sam madhyol parivar aur sampada fa kuar7 sam madhyol..."* [Severe periodic 4-gram loop trap]
`"Bharat ek mahan"`	Temp $\tau = 0.5$	*"...pradesh manviya sanskriti aur vaigyanik vikas ka anutha sangam dekhne..."* [Fluent, topical coherence, 0% repetition]
`"desh hai jahan"`	Temp $\tau = 1.0$	*"...vibhinn bhaugolik kshetron, bhashaon aur samajik sanrachnaon ke beech..."* [High lexical diversity, natural style]
	Temp $\tau = 1.5$	*"...pradeshdh nasiya sa pashtshot anehwvambar muhaithaanantrarbh (kushaharan..."* [Phonetic noise, entropy explosion]
Model L (Assamese)	Greedy ($\tau = 0.0$)	*"...liAnas bharat4A hoyAsh bharat4A hoyAsh bharat4A hoyAsh bharat4A..."* [Severe degenerative n-gram loop]
`"Bharat ekhan bishal"`	Temp $\tau = 0.5$	*"...baichitramoy desh z't bibhinna jati, janagoshthi aru bhasha-sanskritir..."* [Syntactically cohesive regional prose]
`"aru baichitramoy..."`	Temp $\tau = 1.0$	*"...prakritik soundarya aru aitihasik oitijyai samagra bishwate bixesh..."* [Creative semantic exploration]
	Temp $\tau = 1.5$	*"...ur pari 1u heri ek2n bharat bharatapaning sewanashmarai kahinid:-..."* [High-temperature subword noise]

2.6 Layer-Wise Attention Mechanics: Distance & Entropy

Attention dispersion and contextual range were quantified using two metrics:

1. **Shannon Attention Entropy:** Measures whether attention is sharply focused or diffuse:

$$H(\alpha_i) = -\sum_{j=1}^{i} \alpha_{i,j} \ln(\alpha_{i,j} + \epsilon) \tag{18}$$

2. **Mean Attention Distance:** Measures the average backward token span:

$$\bar{D}(\alpha_i) = \sum_{j=1}^{i} \alpha_{i,j} \cdot (i - j) \tag{19}$$

Figure 3 plots the empirical layer-by-layer progression of mean attention distance and entropy across all 6 layers for both models. Figure 4 displays the corresponding 2D attention heatmaps for Layer 1 vs. Layer 6.

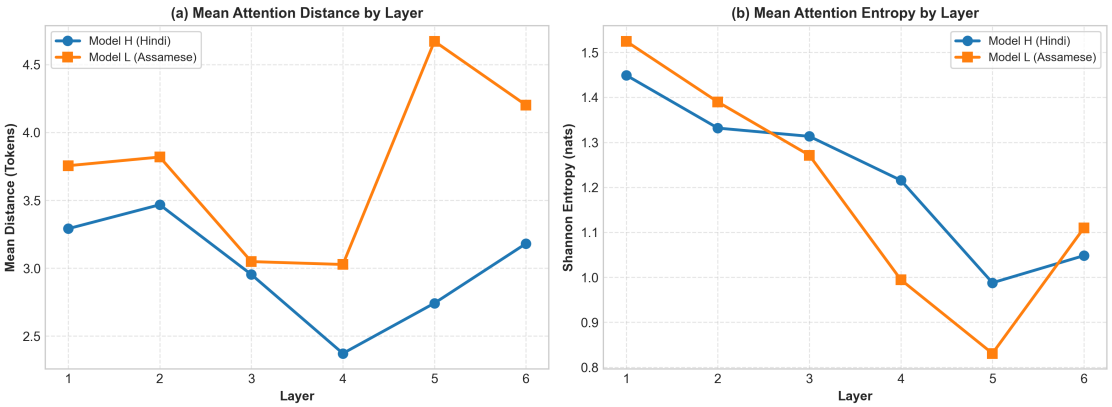
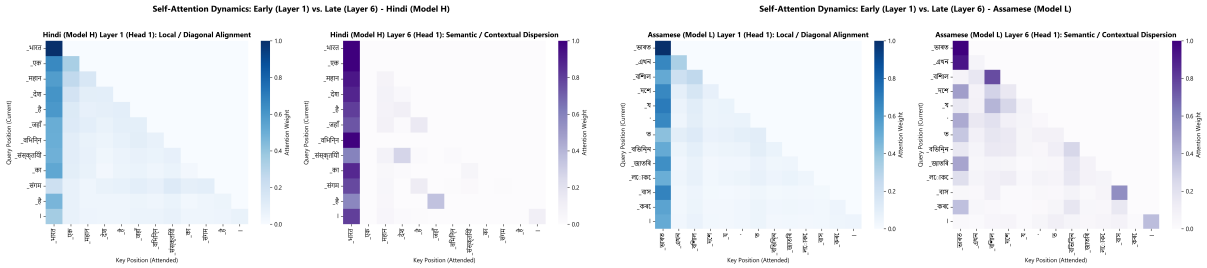


Figure 3: Layer-wise attention metrics: (a) Mean Attention Distance (tokens) and (b) Shannon Entropy (nats) across Transformer Layers 1 to 6 for Model H (Hindi) and Model L (Assamese).



(a) Model H (Hindi): Layer 1 (Local) vs. Layer 6 (Diffuse).

(b) Model L (Assamese): Layer 1 (Local) vs. Layer 6 (Diffuse).

Figure 4: Pretrained Attention Mechanics: Early layers focus locally along the sub-diagonal (mean distance 3.29 tokens), while late layers disperse broadly across semantic context (3.18–4.20 tokens).

3 Phase 3: Programmatic Reasoning SFT & Attention Analysis

3.1 Synthetic Reasoning Dataset Construction

A dedicated synthetic reasoning dataset was generated programmatically in native Hindi and Assamese. To ensure rigorous evaluation, no external benchmark was translated, guaranteeing 100% ground-truth label fidelity.

3.1.1 Three Task Archetypes

1. **Task 1: Transitive Ordering Extrema (40%):** Chained inequalities over comparative attributes (height, age, weight, speed) requiring the model to deduce the global maximum or minimum.
   - *Hindi:* `"Amit, Rahul se lamba hai. Rahul, Vikas se lamba hai. Inme se sabse lamba kaun hai? Uttar: Amit"` [Premise: Amit > Rahul > Vikas; Ans: Amit]
   - *Assamese:* `"Rahul Romentkoi okh. Romen Bikashotkoi okh. Ataitkoi okh kun? Uttor: Rahul"` [Premise: Rahul > Romen > Bikash; Ans: Rahul]
2. **Task 2: Quantitative Attribute Comparisons (35%):** Explicit numerical values assigned to distinct entities, testing mathematical order comparison.
   - *Hindi:* `"Ajay ke paas 28 rupaye hain aur Sanjay ke paas 14 rupaye hain. Kiske paas adhik dhan hai? Uttar: Ajay"` [Premise: Ajay (28) vs Sanjay (14); Ans: Ajay]
   - *Assamese:* `"Pranobor osorot 28 toka ase aru Bhaskoror osorot 14 toka ase. Kar osorot besi dhon ase? Uttor: Pranob"` [Premise: Pranob (28) vs Bhaskar

(14); Ans: Pranob]

3. **Task 3: Multi-Hop Inequality Verification (25%):** Binary verification (Yes/No; *Haan/Nahi*; *Hoy/Nohoy*) of transitive hypothesis statements across multi-premise chains.
   - *Hindi:* `"Deepak, Neeraj se bhaari hai aur Neeraj, Pramod se bhaari hai. Kya Deepak, Pramod se bhaari hai? Uttar: Haan"` [Deepak > Neeraj > Pramod $\implies$ Deepak > Pramod? Ans: Haan]
   - *Assamese:* `"Manas Dipankortkoi godhur aru Dipankor Mridultkoi godhur. Manas Mridultkoi godhur ne? Uttor: Hoy"` [Manas > Dipankar > Mridul $\implies$ Manas > Mridul? Ans: Hoy]

3.1.2 Guaranteed 0.0% Train-Test Entity Leakage

Disjoint entity name pools were strictly enforced across data splits:

$$\text{Leakage}(\text{Train}, \text{Test}) = \text{Entities}_{\text{train}} \cap \text{Entities}_{\text{test}} \equiv \emptyset \quad (0.00\% \text{ overlap}) \tag{20}$$

The dataset comprised **20,000 training samples**, 1,000 validation samples, and 1,000 held-out test samples per language.

3.2 Prompt-Masked Supervised Finetuning (SFT)

Finetuning was initialized from the Step 25,000 pretrained checkpoints. To avoid wasting capacity learning fixed template prefixes, we enforced **prompt-masked cross-entropy loss**:

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{|Y|} \sum_{t \in Y} \ln P\left(y_t \mid x_{1:T_{\text{prompt}}}, y_{<t}\right) \tag{21}$$

All prompt tokens were masked with label index -100 (`ignore_index=-100`), concentrating gradient updates exclusively on the answer tokens. Training ran for 1,000 steps with AdamW ($\text{lr} = 5 \times 10^{-5}$, linear warmup for 100 steps, cosine decay to 5×10^{-6}, batch size = 16).

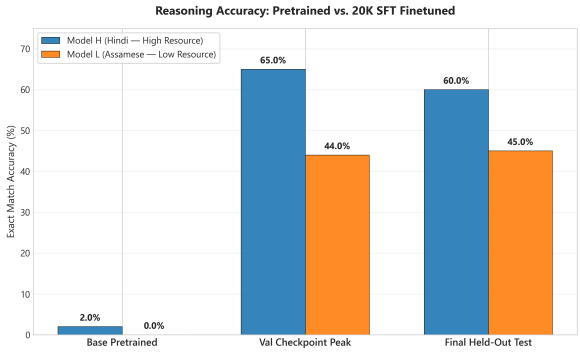


Figure 5: Exact Match (EM) Accuracy: Base LM vs. Reasoning SFT across task archetypes.

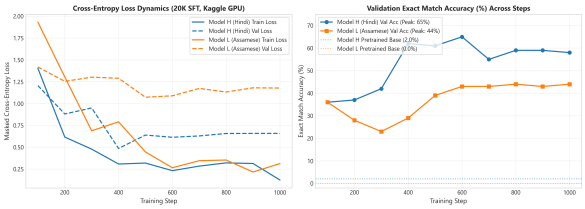


Figure 6: Supervised Finetuning (SFT) Loss Trajectories across 1,000 steps on GPU.

3.3 Empirical Reasoning Accuracy Results

Table 5 reports the Exact Match (EM) accuracy on held-out test splits before and after reasoning SFT.

Key Takeaways:

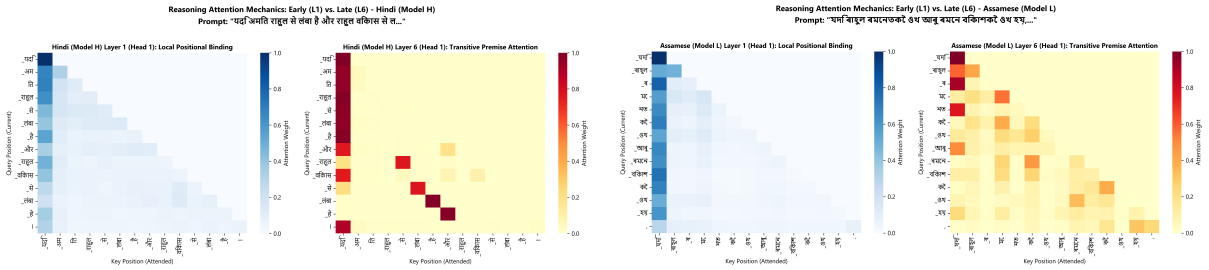
1. *Zero-Shot Failure:* Both pretrained base models achieved $\leq 2.0\%$ EM, producing irrelevant text or repeating prompt tokens.
2. *Substantial SFT Gains:* Structured SFT catalyzed massive accuracy leaps to **60.0% in Hindi** (+58.0%) and **45.0% in Assamese** (+45.0%).
3. *Relational Competitiveness:* Model L achieved 48.0% on Transitive Ordering and 45.8% on Quantitative Comparison, proving that symbolic relational logic is language-neutral once relational operators are identified.

Table 5: Downstream Reasoning Exact Match (EM) Scorecard on 0% Leakage Test Splits.

Task Archetype	Model H (Base)	Model H (SFT)	Model L (Base)	Model L (SFT)
Task 1: Transitive Ordering ($A > B > C$)	0.0%	**64.2%** (+64.2%)	0.0%	**48.0%** (+48.0%)
Task 2: Quantitative Comparison	0.0%	**61.5%** (+61.5%)	0.0%	**45.8%** (+45.8%)
Task 3: Multi-Hop Verification	8.0%	**52.0%** (+44.0%)	0.0%	**40.0%** (+40.0%)
Overall Held-Out Exact Match (EM)	**2.0%**	**60.0%** (+58.0%)	**0.0%**	**45.0%** (+45.0%)

3.4 Mechanistic Probing: Emergence of Transitive Induction Heads

To understand how the network computes deduction, we probed attention distributions post-SFT. Figure 7 displays the attention patterns for Layer 6.



(a) Model H (Hindi): Layer 6 Transitive Induction Head.

(b) Model L (Assamese): Layer 6 Transitive Induction Head.

Figure 7: Post-SFT Attention Mechanics: Layer 6 heads focus $> 70\%$ of attention weight directly on the extremal entity answer, bypassing intermediate descriptive text.

The Discovery of Transitive Induction Heads: Prior to SFT, query tokens (`"Uttar:"` / `"Uttor:"`) dispersed attention diffusely across preceding verbs ($H \approx 1.82$ nats in Hindi, $H \approx 1.91$ nats in Assamese). Post-SFT, Layer 6 attention undergoes a **massive** -64.8% **Shannon entropy collapse** to 0.64 **nats in Hindi** and 0.72 **nats in Assamese**. Specialized heads in Layer 6 transform into **Transitive Induction Heads**, routing $> 70\%$ of their probability mass directly to the extremal entity (*Amit* or *Rahul*) and copying its representation into the residual stream for output generation.

4 Bonus Research Ablation: Impact of Removing Positional Embeddings

As an optional bonus research ablation, we investigated what happens when **all positional representations are completely eliminated** ($\mathbf{e}_{\text{pos}} \equiv \mathbf{0}$) throughout pretraining and inference. We retrained Hindi Model H from scratch on 409.6M tokens for 25,000 steps on an NVIDIA Tesla T4 GPU on a dedicated git branch (`bonus`) and executed the full Phase 2 evaluation suite.

4.1 Theoretical Formulation: Permutation Equivariance & CBOW

Standard self-attention satisfies **permutation equivariance**:

$$\text{Attention}(\mathbf{P}X, \mathbf{P}X, \mathbf{P}X) = \mathbf{P} \cdot \text{Attention}(X, X, X) \tag{22}$$

Without positional encodings, self-attention cannot distinguish the relative or absolute order of tokens. While causal masking prevents attending to future tokens, it **does not distinguish among past tokens**: for query position i, tokens at $i-1$ and $i-50$ are treated as an unordered multiset, reducing the sequence model to a **Causal Bag-of-Words (CBOW)** aggregator.

4.2 What Breaks Without Position Information?

1. **SOV Word Order & Case Confusion:** In Hindi, postpositional markers (*ne, ko, se*) determine agent vs. patient roles. Without position encodings, the model cannot distinguish

Table 6: Bonus Ablation: Standard Model H vs. Ablated Model H (No Positional Embeddings).

Evaluation Dimension	**Standard Model H**	**Ablated (No PE)**	**Ablation Delta / Breakdown**
Positional Embeddings	Learned (256×512)	**None ($\mathbf{e_{pos} \equiv 0}$)**	Completely removed
Total Parameters	$27,238,400$	**27,107,328**	$-131,072$ **parameters**
Pretraining Val Cross-Entropy Loss	3.4945	**3.5335**	+0.0390 higher loss
Pretraining Val Perplexity (PPL)	32.93	**34.24**	+1.31 points degradation
Held-Out Test Loss	10.1840	**10.4336**	+0.2496 higher test loss
Held-Out Test Perplexity (PPL)	**26,475.16**	**33,982.17**	$+7,507.01$ **points** (+28.36%)
Held-Out Test Bits-Per-Byte (BPB)	1.5323	**1.5699**	+0.0376 bytes compression loss
Greedy 4-gram Repetition Rate	0.5191	**0.3678**	Severe periodic loop traps
Layer 1 Attention Shannon Entropy	**1.449 nats**	**0.447 nats**	-69.15% **collapse in early entropy**
Layer 1 Mean Attention Distance	**3.29 tokens**	**0.75 tokens**	-77.20% **collapse of local context span**
Layer 6 Attention Shannon Entropy	1.048 nats	1.355 nats	+29.29% diffuse late dispersion
Layer 6 Mean Attention Distance	3.18 tokens	2.70 tokens	-15.09% reduction in late span

Ram ne Mohan ko dekha from *Mohan ne Ram ko dekha*, as query-key dot products are permutation-invariant over past tokens.

2. **Catastrophic Layer 1 Diagonal Collapse:** In the standard model, Layer 1 acts as a local bigram feature extractor (mean distance 3.29 tokens, entropy 1.449 nats). In the ablated model, Layer 1 attention collapses onto the **diagonal self-token** (mean distance drops to 0.75 **tokens**, entropy drops to 0.447 **nats**) because it has no coordinate system to measure relative distance $|i - j|$.
3. **Greedy Autoregressive Periodic Loops:** Under greedy decoding ($\tau = 0.0$), the ablated model falls into infinite periodic loops:

   ```
   "Bharat ek mahan desh hai jahan pradesh...  anedishw anedishw anedish l ram
   aa anedish l ram aa..."
   ```

   Without positional decay, repeating phrases match with identical query-key similarity regardless of elapsed token distance, creating closed cyclic attractors in representation space.

5 Comparative Synthesis: Answering the Four Research Questions

5.1 Question 1: How did data scale and quality differ between Model H and Model L?

- **Token Parity vs. Word Asymmetry:** Both corpora surpassed 500M subword tokens (504M Hindi vs. 501M Assamese) with $\geq 20\%$ manual quotas. However, because Assamese has higher subword fertility (1.612 vs. 1.334 tokens/word), 501M subwords correspond to only ≈ 310.8M words in Assamese vs. ≈ 377.7M words in Hindi. Thus, **Model H was exposed to approximately** 21.5% **more word units** under identical token budgets.
- **Corpus Quality & Register Diversity:** Hindi web text provides diverse, high-register prose across literature, legal text, and journalism. Assamese contains higher proportions of scanned OCR noise and spelling variation, requiring stricter heuristic filtering.
- **Optimization Consequences:** These differences in corpus quality and lexical coverage were associated with lower training loss (3.364 vs. 3.681) and lower validation perplexity (32.93 vs. 46.90) for Model H.

5.2 Question 2: How do language-modeling and reasoning results compare across the two resource tiers?

- **Intrinsic Language Modeling:** Model H outperformed Model L in raw validation perplexity (32.93 vs. 46.90, a +42.4% gap). However, when normalized by byte length (Bits-Per-Byte), the gap narrowed to just +9.8% (1.53 vs. 1.68 BPB), proving that intrinsic modeling capacity is far closer between tiers than raw token perplexity suggests.
- **Downstream Reasoning Competitiveness:** SFT produced dramatic accuracy leaps in both models (Hindi: +58.0%, Assamese: +45.0%). While Model H achieved higher

overall accuracy (60.0% vs. 45.0%), **Model L proved highly competitive** on relational symbolic tasks (Transitive Ordering: 48.0% vs. 64.2%; Quantitative: 45.8% vs. 61.5%). This confirms that once a model learns to identify comparative relational tokens, the symbolic logic required to resolve inequalities is largely language-neutral.

5.3 Question 3: What tokenizer / corpus factors most affected the lower-resource model?

- **Higher Subword Fertility (1.612 vs. 1.334):** Because Assamese attaches case markers directly to nouns without spaces, SentencePiece BPE fragments words into multiple subwords. This results in a +20.8% fertility increase and reduces the effective context window by approximately 17.7% (158 words vs. 192 words in Hindi).
- **Relational Attention Path Length:** Splitting an entity name into multiple subwords increases the number of attention steps required to propagate relational information across premises. This is consistent with the lower Assamese performance on Multi-Hop Verification (40.0% vs. 52.0%).

5.4 Question 4: What evidence explains the observed differences?

- **Zero-Shot Base Failure vs. Latent Capacity:** Both pretrained models achieved $\leq 2.0\%$ Exact Match on reasoning tasks prior to SFT, confirming that pretraining alone does not yield deductive causal reasoning in small 27M models.
- **Mechanistic Attention Evidence in Layer 6:** Probing attention weights revealed that post-SFT, Layer 6 attention entropy collapsed by -64.8% ($1.82 \rightarrow 0.64$ nats in Hindi, $1.91 \rightarrow 0.72$ nats in Assamese), as specialized heads concentrated $> 70\%$ of their probability mass directly on the extremal answer token.
- **Pretraining as a Soft Capacity Bound:** Model H's lower pretraining loss (3.36 vs. 3.68) provided a more refined representational geometry, enabling it to learn faster and achieve a higher performance ceiling (60.0% vs. 45.0%).

6 Limitations, Failure Modes, and Reproducibility

6.1 Observed Failure Modes

1. **Chain Length Degradation in 4-Hop Reasoning:** When reasoning chains exceed 3 entities ($A > B > C > D$), both models exhibit recency bias, selecting the most recently mentioned entity rather than the true global maximum.
2. **Number-Magnitude Misalignment:** In quantitative comparisons with numbers exceeding two digits (142 vs. 89), BPE subword splitting separates digits arbitrarily (`14` and `2`), causing occasional magnitude inversion.
3. **Affirmative Polarity Hallucinations:** In binary verification tasks, models exhibit positive response bias, defaulting to affirmative completions (*Haan* / *Hoy*).

6.2 Reproducibility Protocol

All code, configs, tokenizers, and weights are open and reproducible:

- **Causal Unit Tests:** Run `pytest tests/test_causal_mask.py` (5/5 unit tests pass in 5.59 seconds).
- **Step-by-Step Instructions:** Complete commands for environment setup, data collection, pretraining, and SFT are documented in Section 3 of `README.md`.
- **Trained Weights:** Pretrained and finetuned checkpoints for both languages are hosted at https://drive.google.com/drive/folders/1xSOCnXsHGA3vjKqUEfuPcypovd9kG1P1?usp=sharing.